

5.3 Homeostasis

Homeostasis involves co-ordination which may be chemical or electrical. The aim of this section is to consider the principles of homeostasis with reference to the control of heart rate, body temperature and blood glucose concentration. The implications of a failure of homeostatic mechanisms are also considered.

Learners gain an understanding of the kidney as an organ of homeostasis and excretion, and the effects of kidney disease, and diabetes, are studied.

The ethics of organ donation and the potential uses of cloning and stem cell technology in treatments are also considered.

Learners are expected to apply knowledge, understanding and other skills gained in this section to new situations and/or to solve related problems.

5.3.1 The principles and importance of homeostasis

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) the general principles of homeostasis in the maintenance of a stable internal environment	To include the role of receptors, effectors and negative feedback AND the concept of normal ranges for temperature, pH, blood glucose concentration and blood pressure. AND contrast positive and negative feedback systems using oxytocin as an example of positive feedback. <i>M1.6</i>
(b) the nervous and hormonal control of heart rate	To include the roles of the sympathetic and parasympathetic nervous system and adrenaline.
(c) the control of body temperature	To include the role of peripheral temperature receptors, the thermoregulatory centre in the hypothalamus and the responses to rising and falling temperatures (sweating, shivering, vasoconstriction and vasodilation).
(d) the regulation of thyroxine release and the effect of thyroxine on metabolic rate	
(e) the techniques for and the importance of measuring core body temperature	To include oral, tympanic (ear), axillary (arm pit) and rectal methods and the significance of readings outside the normal range in adults and children. <i>M1.1, M1.2, M1.6</i> PAG10 HSW3, HSW4, HSW5, HSW6

(f) the causes, symptoms and treatment of hypothermia and hyperthermia.

To include the consequences of 'fuel poverty' and analysis of data showing correlations between weather conditions and incidence of both hypothermia and hyperthermia with respect to climate change.

M0.3, M1.7

HSW9, HSW10, HSW11, HSW12